8 In this question you must show detailed reasoning.

A is the point (1, 0), B is the point (1, 1) and D is the point where the tangent to the curve $y = x^3$ at B crosses the x-axis, as shown in Fig. 8. The tangent meets the y-axis at E.

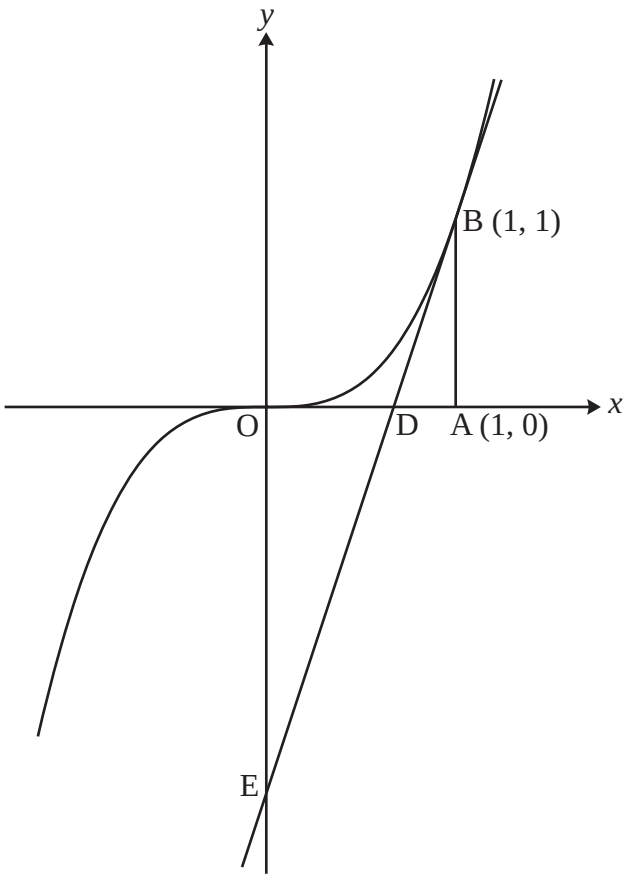

Fig. 8

(a) Find the area of triangle ODE. **[6]**

(b) Find the area of the region bounded by the curve $y = x^3$, the tangent at B and the y-axis. **[4]**